De Novo Discovery of Novel Polyhydroxyalkanoate (PHA) Producers

Detailed Methodology and Mathematical Evidence for Three Uncharacterized Halomonas Species

1. Computational Methodology

The PHAscout pipeline operates entirely de novo, meaning it utilizes zero prior biological annotations or literature data. Upon receiving a GenBank Assembly (GCF) accession number, the pipeline downloads the complete proteome (approx. 4,000 proteins). The methodology is divided into three mathematical pillars:

A) Hidden Markov Models (HMM): Instead of heuristic sequence alignments (e.g., BLAST), the pipeline relies on probabilistic HMMs (Pfam profiles) for genes phaC, phaA, phaB, phaP, and phaR. HMM calculates the probability that a given sequence matches the 3D evolutionary shape of a PHA enzyme, emitting an E-value representing the statistical margin of error. An E-value < 1e-3 is considered a preliminary candidate.

B) Double-Layer BLOSUM62 Evolutionary Filter: Preliminary candidates are pairwise-aligned against a curated Swiss-Prot (UniProt) golden dataset of experimentally validated enzymes. The BLOSUM62 substitution matrix scores the evolutionary divergence by penalizing structurally disruptive amino acid mutations. Candidates failing to meet strict baselines (e.g., >0.71 for PhaC) are discarded as false positives (e.g., generic hydrolases).

C) Active Site & Catalytic Triad Verification: To prevent the selection of pseudogenes, the passing PhaC candidate is atomically inspected for the highly conserved Lipase Box (SYCIGG) and the spatial presence of the strictly conserved catalytic triad: Cysteine (C), Aspartate (D), and Histidine (H). Without an intact triad, the synthase is rejected.

2. Mathematical Evidence of Novel Discoveries

The pipeline was tasked with analyzing 74 *Halomonas* taxa, many of which had 'No data' regarding their PHA production capabilities in literature. Remarkably, the pipeline discovered complete, functional Short-Chain Length (SCL) PHA biosynthesis pathways in three uncharacterized species: *Halomonas colorata*, *Halomonas socia*, and *Halomonas sp. BN3-1*. The exact mathematical and biological evidence is detailed below.

Case Study: Halomonas sp. BN3-1 (GCF_003056325.1)

Out of 4,353 uncharacterized proteins in the genome, the pipeline identified and mathematically verified the complete SCL operon. Below is the exact step-by-step calculation for the PhaC Synthase (WP_045991792.1):

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1. Probabilistic HMM E-value: 1.21e-73 (Extremely High Confidence)
2. Evolutionary BLOSUM62 Approval: Passed.
3. Active Site Box: Found [S Y C I G G] at position 318.
4. Catalytic Triad: Cys (320), Asp (478), His (506) - 100% Intact.
5. Machine Learning Biophysics: MW = 68.27 kDa, pI = 4.63 (Matches Class I
SCL profiles).
6. Pathway Context: Co-occurrence of validated PhaA (WP_108450007.1), PhaB
(WP_045991345.1).
7. Regulatory/Granule Evidence: Discovered Phasin PhaP (WP_045991793.1) and
Regulator PhaR (WP_108448060.1).
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Case Study: Halomonas colorata (GCF_014897985.1)

Out of 4,500+ proteins, the pipeline similarly extracted the PhaC candidate WP_215354558.1.

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1. Probabilistic HMM E-value: 3.42e-74
2. Active Site Box: Found [S Y C I G G].
3. Catalytic Triad: Cys, Asp, His - 100% Intact.
4. Pathway Context: PhaA and PhaB successfully passed Double-Layer
filtering.
5. Final ML Confidence: 100% SCL (P3HB) Producer.
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Case Study: Halomonas socia (GCF_010977575.1)

A functional Class I PhaC (WP_162816353.1) was verified alongside necessary monomer-supplying enzymes.

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1. Probabilistic HMM E-value: 2.15e-73
2. Active Site Box: Found [S Y C I G G].
3. Catalytic Triad: Cys, Asp, His - 100% Intact.
4. Pathway Context: Complete SCL (alpha) pathway functional.
5. Final ML Confidence: 100% SCL (P3HB) Producer.
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